

THE FOUR PILLARS OF MODERN TECHNOLOGY

Cloud Computing, AI, IoT & Big Data

How connected infrastructure, intelligent systems, and data at scale are reshaping business and everyday life

A Technology Overview



What We'll Cover



01

Cloud Computing

Service & deployment models that host modern apps



02

Artificial Intelligence

How machines learn, reason & automate



03

Internet of Things

Connected devices sensing the physical world



04

Big Data

Turning massive datasets into real value

What Is Cloud Computing?

Cloud computing delivers computing power, storage, and software over the internet — on demand, without owning physical hardware.

Providers offer different levels of managed infrastructure, letting businesses choose how much they build versus how much they rent.



IaaS

Infrastructure as a Service — raw compute, storage & networking



PaaS

Platform as a Service — ready-to-use dev & deployment tools



SaaS

Software as a Service — finished applications over the web



Serverless

Code runs on demand with no server management at all

Cloud Deployment Models



Public

Shared infrastructure hosted by a third-party provider, e.g. AWS, Azure.



Private

Dedicated infrastructure used by a single organization for control.



Hybrid

A mix of public and private, balancing flexibility and security.



Multi-Cloud

Multiple providers used together to avoid lock-in and add resilience.

What Is Artificial Intelligence?

AI refers to systems that perform tasks normally requiring human intelligence — recognizing patterns, understanding language, and making predictions.

Modern AI is trained on large datasets, often drawing on the same cloud and data infrastructure covered elsewhere in this deck.



Machine Learning



Natural Language



Computer Vision



**Robotics &
Automation**

AI in Everyday Business



Process Automation

AI handles repetitive tasks like data entry, scheduling, and document review.



Personalization

Recommendation engines tailor content and products to individual users.



Predictive Analytics

Models forecast demand, risk, and customer behavior ahead of time.



Conversational AI

Chatbots and virtual assistants handle support and routine queries.

What Is the Internet of Things?

IoT connects everyday physical devices — sensors, machines, appliances — to the internet so they can collect and exchange data automatically.

These devices typically send data to the cloud for storage and analysis, closing the loop between the physical and digital worlds.



Sensors

Capture real-world signals like temperature or motion



Connectivity

Wi-Fi, cellular & Bluetooth move data to the network



Edge Devices

Process data locally before sending it onward



Cloud Platform

Stores, analyzes & acts on incoming device data

IoT Applications Across Industries



Smart Homes

Thermostats, lighting & security systems controlled remotely.



Wearables

Fitness trackers and smartwatches monitor health in real time.



Industrial IoT

Sensors on factory equipment predict maintenance needs early.



Smart Cities

Connected traffic, utilities & public services improve efficiency.

The Five V's of Big Data



Volume

Massive amounts of data generated every second



Velocity

The speed at which new data is created & processed



Variety

Structured, unstructured & semi-structured formats



Veracity

Accuracy & trustworthiness of the data collected



Value

The insight & business benefit data ultimately delivers

How Organizations Handle Big Data



Distributed Processing

Frameworks like Hadoop split huge workloads across many machines.



In-Memory Analytics

Tools like Spark process data in memory for near real-time speed.



Data Warehouses

Structured, query-ready storage built for business reporting.



Data Lakes

Raw, flexible storage that holds data in its native format.

Shared Challenges Across All Four



Security & Privacy

More connected data means a larger attack surface to defend.



Rising Costs

Cloud usage, AI compute, and data storage scale quickly with demand.



Integration Complexity

Connecting cloud, AI, IoT & data systems requires careful architecture.



Talent & Skills Gap

Demand for specialized engineers outpaces the available talent pool.

Key Takeaways

- 1 Cloud computing provides the on-demand infrastructure that powers modern applications.
- 2 AI turns data and compute into automated decisions, predictions, and interactions.
- 3 IoT extends digital systems into the physical world through connected devices.
- 4 Big Data supplies the fuel — the scale and variety of information these systems run on.
- 5 Together, the four pillars reinforce each other more than they operate independently.



Thank You

Questions & Discussion